2021 Caldor Fire Reconstruction against GOFER Hourly Perimeters

1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the Caldor Fire (2021) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 626 h, from the first GOES fire detection at 15 August 2021 at 01:00 UTC to the last GOFER perimeter at 10 September 2021 at 03:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. No model parameter was tuned against the GOFER observation of this fire; the spotting configuration was revised after comparisons for the Kincade and CZU Lightning Complex fires.

2 Reference event, experiment, and observations

GOFER records the Caldor Fire from 15 August 2021 at 01:00 UTC to 10 September 2021 at 03:00 UTC with 620 hourly perimeters, a final perimeter area of 994 km^2 and an official size of 221,835 ac, and 1 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of 0.75 ± 0.21 km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 620 hourly perimeters of this fire on a 51.3-m grid in UTM zone 10N (EPSG:32610). The raster covers 995 km^2, consistent with the final GOFER perimeter area of 994 km^2; arrival times range from 7 to 626 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

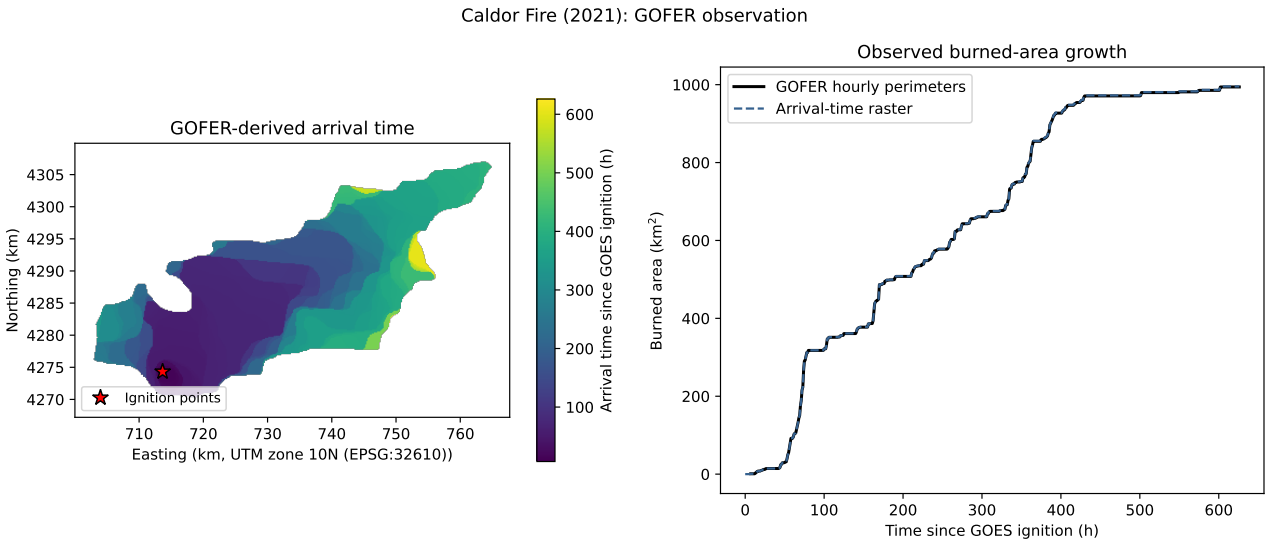


Figure 1: GOFER observation of the Caldor Fire: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

3 Input data provenance and characterization

Input	Source and version	Resolution, units and processing
Terrain	LANDFIRE elevation, slope and aspect [3]	30 m; m and degrees; UTM zone 10N (EPSG:32610)
Surface fuels	LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.2.0	30 m categorical
Canopy	LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.2.0	30 m; percent, m, m and kg m^{-3}
Weather	Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6]	600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees
Fuel moisture	Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather	600 m hourly; percent; derivation method not documented in the available case materials
Ignition	GOFER ignition points [2]	Points with hourly time stamps
Adjustment and initial state	Spread-rate adjustment factor and initial level set	Uniform value of one (no adjustment, nothing burned at the start)

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 2,800 by 1,933 cells of 30 m (4,871 km^2) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from 302 to 3,312 m (median 1,722 m) and slope has a median of 13° and a 95th percentile of 32°; the circular-mean aspect is 247°. By area the domain is 3.22% grass, 12.6% grass-shrub, 9.19% shrub, 49.2%

timber-understory, 14.4% timber litter, 0.0394% slash-blowdown and 11.4% non-burnable fuel models. Canopy cover has a median of 45% and canopy base height a median of 0.9 m.

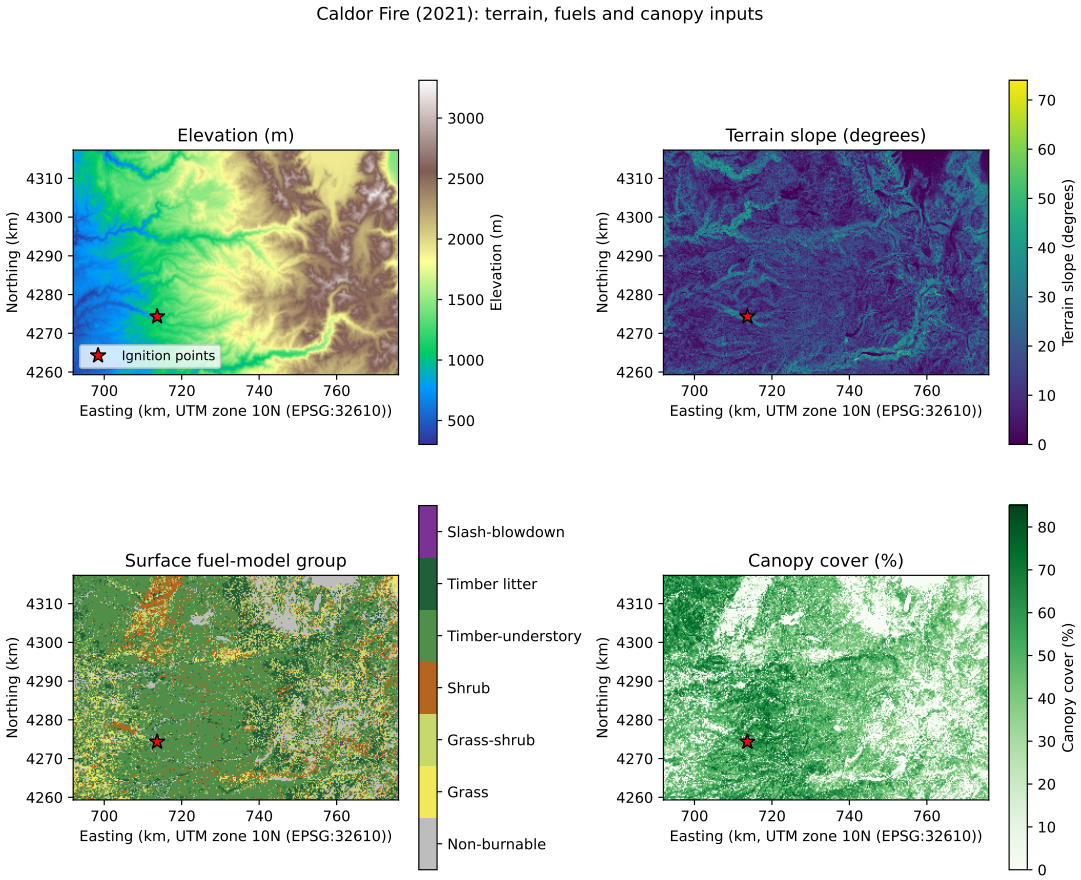


Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

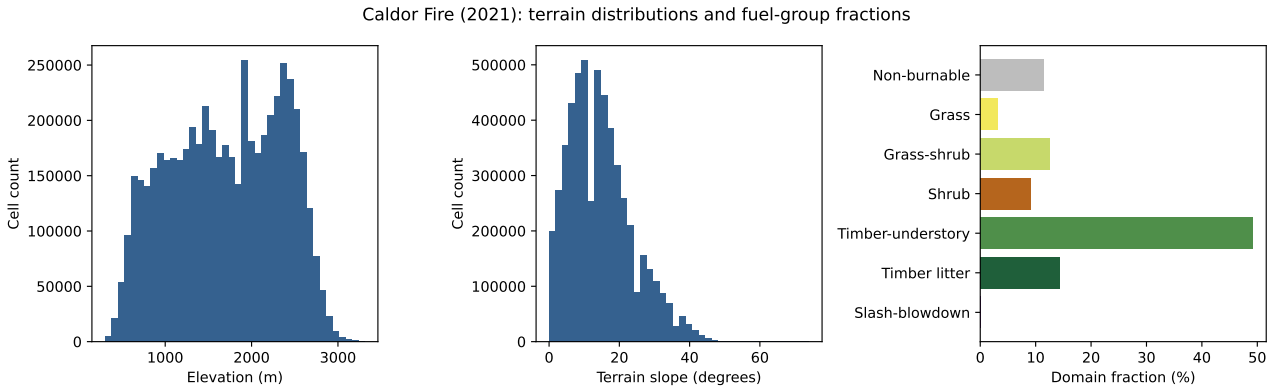


Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 621 hourly fields. Over all hours and cells the 20-ft wind speed has a median

of 3.89 mph and a 95th percentile of 11.4 mph; the circular-mean direction the wind blows from is 206° with a mean resultant length of 0.193 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 5.74% (5th percentile 3.2%), and live herbaceous and woody moisture have medians of 30% and 60%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

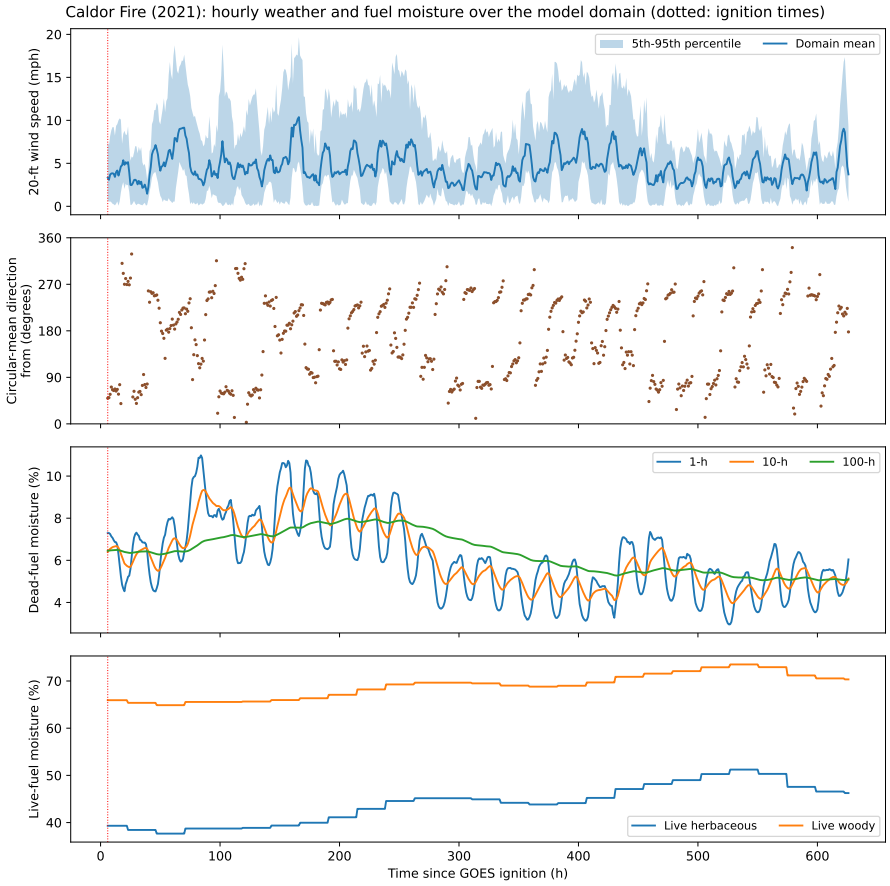


Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

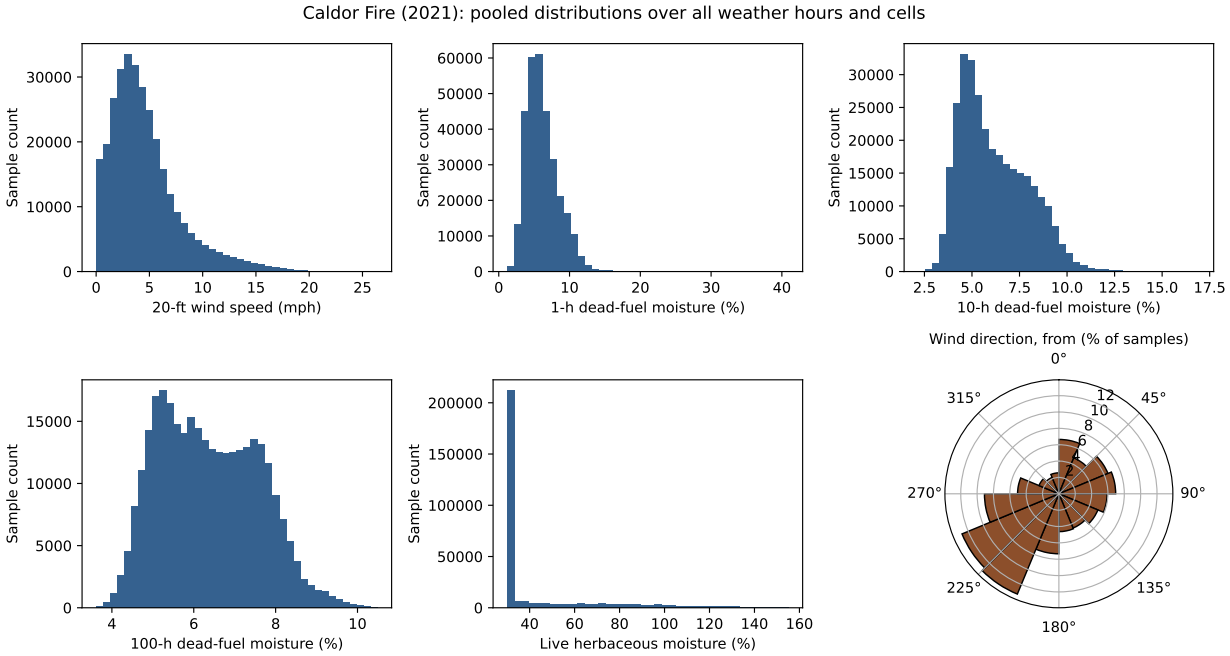


Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 620 h, to the last GOFER perimeter, with a single point ignition. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft min^{-1}, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

Choice	Basis	Evidence and limitation
Ignition points and times	documented	GOFER ignition record; no point required moving
Simulation start and stop	documented	First GOFER ignition point and last GOFER perimeter
Fuel vintage	derived	Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire
Weather and moisture	user-specified	RTMA analysis through CloudFire, as stated by the case owner
Crown fire	documented	ELMFIRE default crown-fire treatment, set explicitly
Firebrand spotting	user-specified	Enabled with ELMFIRE default spotting settings until calibration
Domain margin	user-specified	10 km beyond the final GOFER perimeter
Single deterministic member	user-specified	No ensemble; stochastic spotting uses a fixed random seed

Table 2: Basis of consequential configuration choices.

5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 6 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. The simulation results reported here were produced with ELMFIRE 1.1 at source revision `5ea7460d514b`, as recorded for the CZU Lightning Complex runs and assumed for the other results produced on the same system.

6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time t the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap J_t summarizes footprint agreement; precision P_t diagnoses commission (simulated burning outside the observation), recall R_t omission (observed burning the simulation misses), F_t balances the two, and the area ratio A_t exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

7 Simulation results and field-level comparison

The current status of this case is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available.

Single-ignition baseline without spotting

This result used a single point ignition at the first GOFER ignition point, firebrand spotting disabled and the ELMFIRE default crown-fire treatment, with the same inputs, stop time and numerical controls as the current configuration. The configuration file used was retained with this result.

NOT EVALUABLE: No time-of-arrival result was found for this simulation.

Current case configuration

This result used the current case configuration described in the model configuration section.

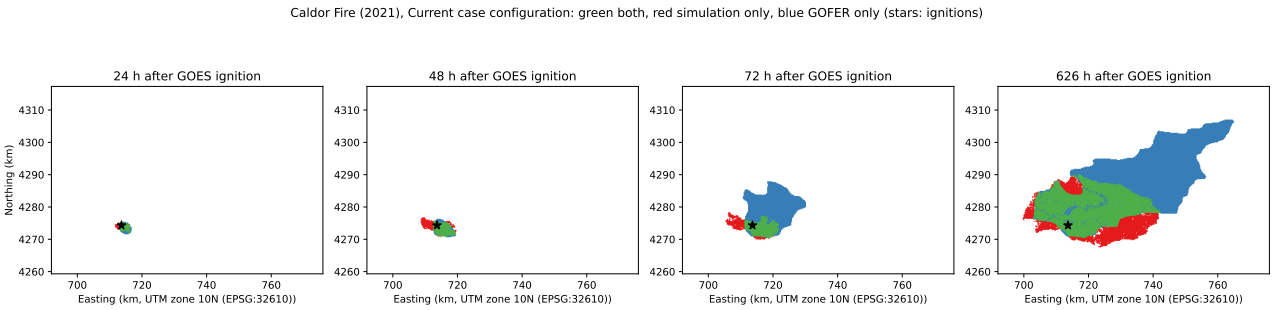


Figure 6: Current case configuration: simulated and GOFER-derived footprints at the fixed comparison times (hours since the GOES ignition hour) on the 30-m model grid: cells burned in both (green), only in the simulation (red), and only in the GOFER observation (blue). Stars mark the configured ignition points.

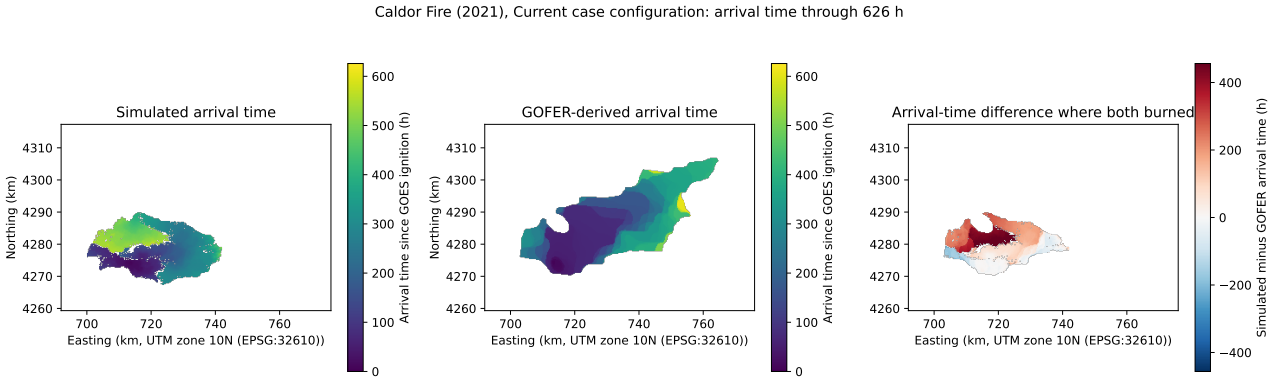


Figure 7: Current case configuration: simulated arrival time, GOFER-derived arrival time and their difference where both burned through the final comparison time, all in hours since the GOES ignition hour. The difference color scale is symmetric and clipped at the 95th percentile of its magnitude.

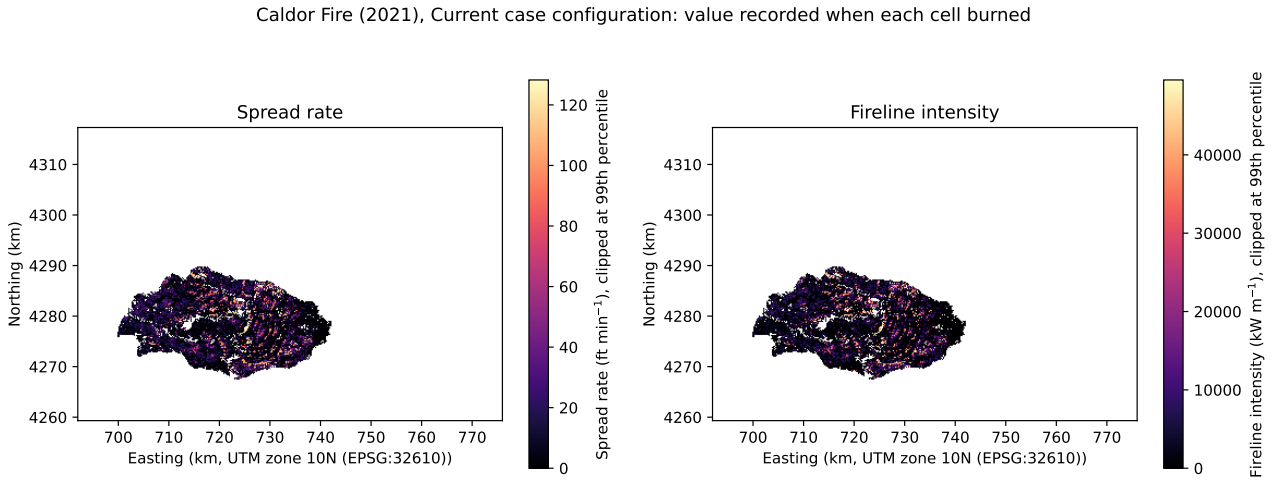


Figure 8: Current case configuration: simulated spread rate and fireline intensity recorded when each cell burned, for the single simulation member at its final output time; color scales are clipped at the 99th percentile.

8 Integrated statistics and calculated metrics

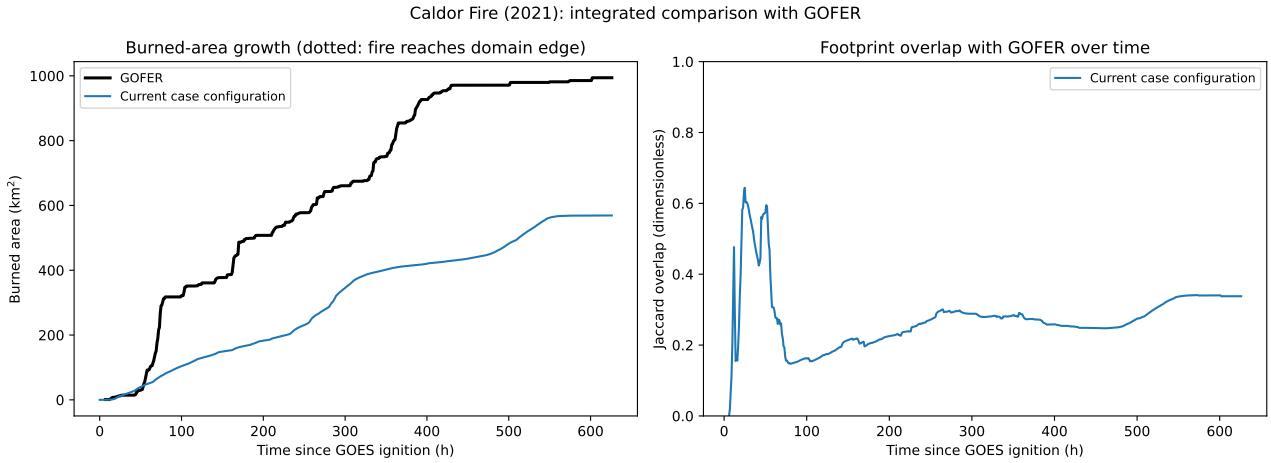


Figure 9: Observed (GOFER) and simulated burned-area growth, and the Jaccard overlap of each simulation with the GOFER footprint, evaluated hourly since the GOES ignition hour. Dotted lines mark the time each simulated fire first reaches the model-domain boundary.

Single-ignition baseline without spotting

NOT EVALUABLE: No time-of-arrival result was found for this simulation.

Current case configuration

Time (h)	GOFER (km^2)	Simulated (km^2)	Area ratio	Jaccard	Precision	Recall	F_1
24	12.1	11.1	0.918	0.63	0.81	0.74	0.77
48	29.2	37.4	1.28	0.57	0.65	0.83	0.73
72	216	68.8	0.319	0.2	0.68	0.22	0.33
626	995	569	0.572	0.34	0.69	0.4	0.51

Table 3: Footprint metrics of the current case configuration against the GOFER-derived footprint at fixed times since the GOES ignition hour, evaluated on the 30-m model grid over valid landscape cells. Time is the comparison instant; areas are burned areas through that instant.

Quantity	Value
Cells burned in both by 626 h	439,052
Mean arrival-time difference, simulated minus GOFER (h)	149
Median arrival-time difference (h)	115
Mean absolute arrival-time difference (h)	172
Root-mean-square arrival-time difference (h)	227
Simulated end since GOES ignition (h)	626
Propagation stalled before the planned end	no
First contact with domain boundary (h)	none
Final simulated area (ac)	140,613
Final crown-fire area (ac)	70,100

Table 4: Arrival-time and integrated statistics of the current case configuration. Arrival-time differences use cells burned in both the simulation and the GOFER observation by the final comparison time.

Precision exceeds recall at the final comparison time, so the simulated footprint lies largely inside the observation but misses much of it: the simulation undergrows or follows a displaced trajectory. The simulated burned area is 0.57 times the observed area at 626 h. Where both burned, the simulated fire arrives on average 148.6 h later than the observed fire (mean absolute difference 172.1 h).

9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the Caldor Fire. Inputs and the observation are characterized in full; the current scientific status is **CHARACTERIZED** with 1 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

References

[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, “Systematically tracking the hourly progression of large wildfires using

GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.